

MODULE 13

SQL Triggers and Automation

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

SQL Triggers and Automation

What is a Trigger?

Auto-executes SQL in response to INSERT, UPDATE, or DELETE.

BEFORE: runs before the operation.

AFTER: runs after the operation.

Uses: audit logging, validation, auto-calculations.

AFTER UPDATE Trigger

```
DELIMITER //
CREATE TRIGGER log_update
AFTER UPDATE ON students FOR EACH ROW
BEGIN
INSERT INTO audit(action, old_v, new_v, ts)
VALUES('UPDATE', OLD.gpa, NEW.gpa, NOW());
END //
DELIMITER ;

-- OLD = values before change
-- NEW = values after change
```

BEFORE Trigger Validation

```
DELIMITER //
CREATE TRIGGER check_salary
BEFORE UPDATE ON employees FOR EACH ROW
BEGIN
IF NEW.salary < OLD.salary * 0.9 THEN
SIGNAL SQLSTATE '45000'
SET MESSAGE_TEXT = 'Cannot cut salary more than 10%';
END IF;
END //
DELIMITER ;
```

INSERT Trigger

```
CREATE TRIGGER after_insert
AFTER INSERT ON students FOR EACH ROW
INSERT INTO profiles(student_id, created_at)
VALUES(NEW.student_id, NOW());
```

Manage Triggers

```
SHOW TRIGGERS;  
SHOW CREATE TRIGGER log_update;  
DROP TRIGGER IF EXISTS log_update;
```

Best practice: keep simple, document behavior,
avoid cascading triggers.